Ilyes Jaouedi

MACHINE LEARNING & SIGNAL PROCESSING FOR RADAR AND CAMERA PERCEPTION

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Paris area, France | **Available for a permanent position (CDI) from mid-2027**

PhD candidate at L2S (CentraleSupélec / Université Paris-Saclay) in an industrial chair with FORVIA. I work from raw radar signals to learned models: high-resolution 3D radar imaging, vision foundation models trained on physics-derived labels, and optimal-transport sensor fusion with statistical guarantees. IEEE ICASSP paper, two patent applications, Google Cloud certified ML engineer.

Experience

PhD Candidate — Camera–Radar Fusion for ADAS *L2S, CentraleSupélec / Univ. Paris-Saclay · FORVIA*
October 2024 – Present Gif-sur-Yvette, France

- **Fast high-resolution 3D radar imaging:** dense range–azimuth–elevation images from raw ADC data on million-voxel grids, validated on a TI IWR6843 + DCA1000 in the FORVIA anechoic chamber and on the 192-element RADIal array. *(Patent application.)*
- **Radar reflectivity from a single image:** fine-tuned Depth Anything V2 with LoRA to predict 77 GHz radar cross-section from one RGB image, trained on physics-derived labels with no human annotation; probed 8 frozen foundation-model backbones. *(Paper in preparation.)*
- **Camera–radar fusion on real driving data:** training-free pipeline on RADIal combining raw radar signals, SAM segmentations and learned reflectivity priors. *(Patent application.)*
- **Sensor fusion with guarantees:** array–camera fusion via unbalanced optimal transport, with Cramér–Rao bounds proving the gain over each sensor alone. *(IEEE ICASSP 2026.)*
- Co-supervised two interns; taught wave-physics tutorials.

Computer Vision Research Intern (final-year project) *CAOR, Mines Paris – PSL*
February 2024 – July 2024 Paris, France

- Reconstructed vehicle trajectories by fusing traffic-camera video with LiDAR; built 3D reconstruction and tracking pipelines for urban traffic scenes.

Computer Vision Engineer Intern *Enova Robotics*
Summer 2023 Sousse, Tunisia

- Implemented and fine-tuned LiDAR–camera fusion models (PV-RCNN, BiProDet) for 3D object detection.

Selected Projects

Radar Foundation Model *Ongoing*
Self-supervised pretraining (masked autoencoders) of a transformer on raw 4D radar cubes, with no lidar, camera or labels, fine-tuned for occupancy, segmentation and ego-motion.

Skills

Radar & signal processing: FMCW, MIMO virtual arrays (TDM, DDM), CFAR, raw ADC processing, TI mmWave, array processing (beamforming, MUSIC, covariance fitting), estimation theory, Cramér–Rao bounds
Machine learning: PyTorch, Hugging Face, vision foundation models (DINOv2, SAM, Depth Anything), LoRA fine-tuning, self-supervised learning, generative models (flow matching, diffusion transformers), 3D object detection
Mathematics: optimal transport (unbalanced, Sinkhorn), convex and non-convex optimisation, sparse recovery
Engineering: Python, C++, MATLAB, SLURM (HPC, A100), Docker, Git, Linux, Google Cloud (Vertex AI, BigQuery; certified Professional ML Engineer)
Languages: Arabic (native), French (B2, DELF), English (professional working proficiency)

Publications & Patents

IEEE ICASSP 2026 **I. Jaouedi**, G. Chardon, J. Picheral. *Sensor Array and Camera Fusion via Unbalanced Optimal Transport for 3D Source Localization. Barcelona. hal-05572212*

GRETSI 2025 **I. Jaouedi**, G. Chardon, J. Picheral. *Fusion Acoustique–Visuelle par Transport Optimal. Strasbourg. hal-05273990*

Patent (FR) 2026 *Two French patent applications with FORVIA and CentraleSupélec (fast 3D radar imaging; camera–radar fusion). Filing in progress.*

AISTATS 2027 *Co-author, sparse training of statistical models (under review); CIFAR-10 experiments.*

Education

PhD, Signal Processing & AI, Université Paris-Saclay (L2S, CentraleSupélec) *2024 – 2027*
Sensor Fusion via Optimal Transport for Driver Assistance. Supervisors: G. Chardon, J. Picheral.

Engineering Degree, Signal Processing & Systems, Polytechnique de Tunisie (ranked 1st, final year) *2021 – 2024*

Preparatory Cycle, Maths & Physics, IPEIM Monastir (ranked 38th /1800 nationally) *2019 – 2021*